Explainable Multilingual Legal Meaning Preservation Using JUDGEBERT

Student Name 1, Student Name 2, Student Name 3
School of Computer Science Engineering and Technology, Bennett University, Greater Noida, India
CSET 346 – Natural Language Processing Project Instructor: Dr. Surabhi Solanki

***Abstract*—Simplifying legal text is useful only if the simplified sentence keeps the legal meaning of the original. JUDGEBERT recently framed this as Legal Meaning Preservation (LMP) for French insurance clauses. We extend LMP to English and Hindi and to a new domain, tenancy law. We build EnHi-LMP, 320 parallel English–Hindi original–simplified pairs with typed legal errors and 320 sanity pairs; LegalDiff, 19 bilingual legal-difference features with a rule-based explainer; and ML-JudgeBERT, a JUDGEBERT-style cross-encoder regressor. Across five experiments, generic similarity correlates weakly with LMP ($r \leq 0.37$) and degrades on Hindi, while LegalDiff-based models reach $r = 0.60$, transfer across languages and domains with small losses, score 0.56–0.59 zero-shot on human-labelled FrJUDGE, pass identical/unrelated sanity checks and detect legal errors with F1 0.79. A cross-encoder trained from scratch is weak alone ($r = 0.12$), motivating pretrained multilingual backbones. Labels in this version are rubric-based silver labels pending double human annotation.**

***Index Terms*—legal meaning preservation, text simplification, JUDGEBERT, Hindi NLP, explainable evaluation, cross-lingual transfer.**

I. Introduction

Plain-language rewriting of contracts, insurance policies and tenancy agreements is increasingly automated with large language models. A rewrite is useful only if it keeps the *legal* meaning of the source: who must do what, under which conditions, with which exceptions, by when and for how much. Small surface edits can silently invert that meaning. Removing the single word "not" turns an exclusion into coverage, replacing "must" by "may" turns a duty into an option, and changing "7 days" into "30 days" moves a deadline. Generic similarity measures are almost blind to such edits because the two sentences still share most of their words and most of their distributional meaning.

Beauchemin et al. [1] formalised this problem as *Legal Meaning Preservation* (LMP), released FrJUDGE, a set of 297 French insurance sentence pairs scored from 1 to 10 by five annotators, and trained JUDGEBERT, a BERT-family regressor that predicts the human LMP score. The paper names its own limitations: one language, one legal domain, a small dataset, sentence-level judgements without context and no out-of-domain evaluation.

This project extends JUDGEBERT along four of those axes and adds explainability. Our contributions are:

1. **EnHi-LMP**, a parallel English–Hindi dataset of 320 original–simplified pairs over two legal domains (insurance and tenancy), each pair tagged with one of ten legal error types, plus 320 held-out identical/unrelated sanity pairs.
2. **LegalDiff**, a bilingual set of 19 interpretable legal-difference features (modality, negation, numbers, durations, parties, conditions, exceptions, scope) and a rule-based explainer that returns an error type and a one-sentence rationale.
3. **ML-JudgeBERT**, a JUDGEBERT-style cross-encoder regressor, optionally fused with LegalDiff, trained with JUDGEBERT's identical/unrelated data augmentation.
4. A five-experiment evaluation: generic similarity vs. LMP models (E1), English vs. Hindi and cross-lingual transfer (E2), cross-domain and FrJUDGE transfer (E3), sanity checks (E4) and error analysis with explanation scoring (E5).

II. Related Work

Surface-overlap metrics. BLEU [2] and chrF [3] measure n-gram and character n-gram overlap with a reference; SARI [4] scores simplification operations against references. All reward copying and are insensitive to the legal effect of a dropped negation or condition.

Embedding-based similarity. Sentence-BERT [5] cosine and BERTScore [6] compare contextual representations. They capture paraphrase well but treat "covers" and "does not cover" as near neighbours (Section III).

Learned meaning preservation. MeaningBERT [7] learns to predict human meaning-preservation judgements for general-domain simplification, and LLM-as-judge methods such as G-Eval [8] prompt a large model for a score and rationale. Neither is calibrated to legal effect, and LLM judges are costly and hard to reproduce.

Legal meaning preservation. JUDGEBERT [1] is, to our knowledge, the only metric trained on human judgements of legal meaning. It is limited to French property-damage insurance and returns a score without explanation. Multilingual encoders such as XLM-R [9] and MuRIL [10] make an Indian-language extension possible.

III. Problem Formulation

Given an original legal sentence o and a simplification s in language $\ell \in \{\text{en}, \text{hi}\}$ from domain $d \in \{\text{insurance}, \text{tenancy}\}$, we learn

$$f(o, s; \theta) \rightarrow \hat{y} \in [1, 10], \tag{1}$$

approximating the gold LMP score y. A decision rule maps the score to three classes,

$$\delta(\hat{y}) = \begin{cases} \text{Preserved} & \hat{y} \geq 7.5 \\ \text{Partially preserved} & 4.5 \leq \hat{y} < 7.5 \\ \text{Not preserved} & \hat{y} < 4.5, \end{cases} \quad (2)$$

and an explainer $g(o, s, \hat{y})$ returns an error type $e \in \mathscr{E}$ ($|\mathscr{E}| = 11$ including NONE) and a rationale. Models are judged by Pearson r and RMSE against y.

Motivating example. For the insurance clause "The policy does not cover damage … caused by wear and tear", a faithful rewrite and a rewrite that drops "not" obtain sentence-embedding cosines of 0.85 and 0.84, although their LMP scores are 10 and 1. The same pair in Hindi obtains 0.92 for both.

IV. The EnHi-LMP Dataset

Source clauses. We wrote 80 source clauses modelled on standard Indian motor/health/property insurance wordings (40) and residential rent agreements (40), and translated each into Hindi, giving parallel EN–HI clause twins.

Simplifications. For every clause and language we wrote one faithful plain-language simplification and one simplification containing exactly one legally significant change from a ten-type taxonomy: NEGATION, MODALITY (obligation↔permission), NUMBER, DATE/time limit, omitted condition (OMIT_COND), removed exception (EXC_REMOVED), SCOPE change, party/ENTITY swap, hallucinated content (HALLUC) and MINOR non-essential omission. This yields $80 \times 2 \times 2 = 320$ pairs (Table I).

Labels. Each pair carries an LMP score on JUDGEBERT's 1–10 scale assigned with a written rubric (9–10 identical legal effect; 6–7 minor detail lost; 4–5 secondary condition or scope shifted; 2–3 core condition, exception, figure, party or modality changed; 1 reversal). *In this version the scores are rubric-based silver labels*; the protocol for the final version is independent double annotation, adjudication of differences of three or more points and reporting of Krippendorff's α [11].

Sanity sets. Following JUDGEBERT, each original is paired with itself (gold 10) and with an unrelated simplification from the other domain in the same language (gold 1). These 320 pairs are never used for training.

Table I
EnHi-LMP dataset statistics

	Insurance	Tenancy	Total
English pairs	80	80	160
Hindi pairs	80	80	160
Faithful / single-error	80 / 80	80 / 80	160 / 160
Preserved / Partial / Not (gold)	–	–	160 / 20 / 140
Sanity: identical / unrelated	–	–	160 / 160

Error types (EN+HI): scope 22, omitted condition 20, date 20, modality 16, minor 16, hallucination 16, party 16, negation 14, exception 12, number 8.

V. Methods

A. Generic similarity baselines

We use chrF [3], sentence-embedding cosine and a BERTScore-style [6] greedy token-alignment F1. Embeddings come from WordLlama, a static distillation of Llama-2 token embeddings. Each similarity is mapped to the 1–10 scale by a linear calibrator fitted on the training fold only, so RMSE is comparable across systems.

B. LegalDiff features and explainer

LegalDiff compares o and s with bilingual lexicons and number/duration parsers (Devanagari digits and Indian number formats such as "15,00,000" and "15 lakh" are normalised). Its 19 features are: missing/added numbers, duration mismatches, negation difference, obligation and permission marker differences, a modal-flip flag, lost/added condition and exception markers, quantifier difference, party-role Jaccard distance and subject change, length ratio, lexical coverage, novelty and time-word change. A priority-ordered rule set turns the features and the predicted score into an error type and a templated rationale (e.g. "Time limit changed: 7 → 30").

C. ML-JudgeBERT

Following JUDGEBERT, the pair is cross-encoded as `[CLS]` o `[SEP]` s `[SEP]` with segment embeddings. A Transformer encoder produces token states; we pool h_{CLS} and the mean states u, v of the two segments into $[h_{\text{CLS}}; u-v; u \odot v; |u-v|]$, optionally concatenate the standardised LegalDiff vector, and predict $\hat{y} = 1 + 9\,\sigma(\text{MLP}(\cdot))$. Training minimises MSE with AdamW and a one-cycle schedule, using JUDGEBERT's augmentation: for every training original we add the identical pair (10) and one unrelated pair (1).

Compute constraint. The development environment could not download pretrained checkpoints, so the encoder used here is a 2-layer, 128-dimensional Transformer with a 3k-token BPE vocabulary shared by English and Hindi, trained from scratch on CPU. A pretrained multilingual backbone (XLM-R or MuRIL) is the intended configuration and is discussed in Section IX.

D. LegalDiff-GBR

As an ablation without any text encoder we fit a gradient-boosted regressor (200 trees, depth 3) on the LegalDiff features, with the same augmentation.

VI. Experimental Setup

Protocol. E1 uses 5-fold cross-validation *grouped by clause*: the English and Hindi twins of a clause, and both of its simplifications, always fall in the same fold, so no test clause is seen in training in either language. Learned models are run with three seeds; we report the mean and standard deviation of each metric computed over the pooled out-of-fold predictions. E2 splits the clauses into two halves and trains on one language of one half while testing on the other language of the other half (and the in-language controls). E3 trains on one domain and tests on the other, and additionally evaluates on the official FrJUDGE test split (90 French pairs with *human* LMP labels, mean of five annotators), either zero-shot from EnHi-LMP or after training

Table II
E1: 5-FOLD GROUPED CV ON ENHI-LMP ($n = 320$)

System	$r\uparrow$	$\rho\uparrow$	RMSE↓	MAE↓	Dec.F1↑	NP-F1↑
chrF	0.37	0.36	3.38	3.00	0.28	0.25
Embedding cosine	0.18	0.16	3.58	3.38	0.06	0.06
Token-align. F1	0.18	0.19	3.58	3.39	0.04	0.00
LegalDiff-GBR	**0.60** $\pm$**0.01**	**0.58**	**2.94**	**2.24**	**0.53**	**0.61**
ML-JudgeBERT (text)	0.12 $\pm$0.01	0.08	4.50	3.45	0.35	0.37
ML-JudgeBERT+LegalDiff	0.44 $\pm$0.02	0.45	3.82	2.63	0.42	0.46

Table III
PEARSON r BY LANGUAGE AND DOMAIN (E1/E2)

System	EN	HI	Insurance	Tenancy
chrF	0.36	0.38	0.36	0.38
Embedding cosine	0.38	0.21	0.17	0.19
Token-align. F1	0.42	0.28	0.13	0.22
LegalDiff-GBR	0.60	0.59	0.59	0.61
ML-JudgeBERT (text)	0.10	0.13	0.10	0.13
ML-JudgeBERT+LegalDiff	0.41	0.46	0.36	0.51

on FrJUDGE train+dev (207 pairs). E4 scores the 320 held-out sanity pairs with the E1 models, including a hybrid model trained without augmentation. E5 applies the explainer to the out-of-fold scores of the best model.

Metrics. Pearson r, Spearman ρ, RMSE and MAE against the gold LMP; macro-F1 of the three-way decision; F1 of the "Not preserved" class; and precision, recall and F1 of the error-type classifier.

VII. RESULTS

A. E1: Generic similarity vs. LMP models

Table II shows that the best generic metric is chrF ($r = 0.37$); embedding cosine and token alignment reach only 0.18. Because these similarities vary little between faithful and corrupted rewrites, their calibrated predictions cluster in the middle of the scale and almost never cross the "Not preserved" threshold (NP-F1 ≤ 0.25). LegalDiff-GBR is the best system on every metric ($r = 0.60$, RMSE 2.94, NP-F1 0.61). The JUDGEBERT-style cross-encoder trained from scratch does not learn a useful signal from 256 training pairs on its own ($r = 0.12$); fusing LegalDiff raises it to 0.44. H1 is therefore supported for legal-aware modelling, but not yet for the encoder alone.

B. E2: English vs. Hindi

Table III breaks E1 down by language and domain. The generic embedding metrics lose almost half of their correlation on Hindi (cosine 0.38 $\rightarrow$ 0.21; token alignment 0.42 $\rightarrow$ 0.28): the English-centric tokenizer splits Devanagari into characters, so unrelated Hindi sentences look similar. LegalDiff-GBR is language-stable (0.60 vs. 0.59). In cross-lingual transfer (Table IV) LegalDiff-GBR trained on English reaches $r = 0.48$ on Hindi and, trained on Hindi, 0.54 on English, close to the in-language controls (0.50 and 0.55), whereas embedding cosine collapses to 0.04 on Hindi.

C. E3: New domain and out-of-language transfer

Training on insurance and testing on tenancy, or the reverse, costs LegalDiff-GBR at most 0.06 in r relative to in-domain cross-validation (Tables III–IV). The most informative result is on FrJUDGE, whose labels are genuine human judgements: models trained *only* on EnHi-LMP reach $r = 0.56$ (LegalDiff-GBR) and 0.59 (ML-JudgeBERT+LegalDiff), above chrF (0.47), while embedding cosine is negatively correlated (-0.17). Trained on FrJUDGE itself, LegalDiff-GBR reaches $r = 0.60$ with RMSE 1.96. Note that LegalDiff includes a small French lexicon (Section IX).

D. E4: Sanity checks

Table V reports mean predictions on identical (gold 10) and unrelated (gold 1) pairs. Models trained with augmentation give identical pairs ≈ 10 and unrelated pairs 1.2–1.7; chrF behaves correctly on identical pairs but rates unrelated pairs 2.9. Embedding similarity fails both checks after calibration and gives unrelated *Hindi* pairs $6.6/10$. The text-only encoder places unrelated pairs at 4.8. Removing augmentation from the hybrid model changes little because LegalDiff already separates these extremes.

E. E5: Error analysis and explanations

Fig. 1 shows mean predictions per gold error type. Embedding cosine assigns roughly 6–7 to every type, i.e. it cannot separate a faithful rewrite from a negation, a changed number or a swapped party. LegalDiff-GBR tracks the gold profile, with the clearest separation for negation, numbers, dates, exceptions and omitted conditions, and the weakest for scope changes and hallucinated content, which have no dedicated lexical marker.

Using the out-of-fold LegalDiff-GBR scores, the explainer detects that a legal error is present with precision 0.74, recall 0.85 and F1 0.79. It names the exact error type for 40% of all corrupted pairs (47% of those it flags), with an 11-class macro-F1 of 0.41 (Table VI). Exceptions (0.67), dates (0.58), parties (0.51), numbers (0.50) and modality (0.49) are explained best; scope changes are never typed correctly.

Table VII lists 18 legally difficult cases: the 16 error pairs that embedding similarity over-scores most (at most one per type and language) and two faithful paraphrases with low lexical overlap. Typical successes are polarity reversals expressed with नहीं in Hindi (case 3) and figure or deadline changes (cases 12–13). Typical failures are (i) polarity changed through the verb rather than a negator ("does not cover" $\rightarrow$ "pays for", case 7), (ii) party swaps that keep the same role words (case 6) and (iii) a faithful Hindi paraphrase that reorders the clause and is wrongly flagged (case 17).

VIII. DISCUSSION

Does the model capture legal meaning better than generic similarity? Yes, when legal differences are modelled explicitly. LegalDiff-GBR beats every generic metric in every setting we tested: overall, per language, across domains and on French human judgements, and on decision F1 and sanity checks. The hybrid model beats them overall and on FrJUDGE, but not in

Table IV
E2/E3: TRANSFER ACROSS LANGUAGE, DOMAIN AND TO HUMAN-LABELLED FRJUDGE (PEARSON r)

Train → test	chrF	Emb. cosine	Token-align. F1	LegalDiff-GBR	ML-JudgeBERT (text)	ML-JudgeBERT+LegalDiff
EN→HI	0.39	0.04	0.16	**0.48**	0.08	0.36
HI→EN	0.37	0.26	0.24	**0.54**	0.06	0.42
EN→EN	0.37	0.35	0.42	**0.55**	0.11	0.39
HI→HI	0.37	0.25	0.28	**0.50**	0.14	0.34
Ins.→Ten.	0.40	0.20	0.22	**0.55**	0.19	0.51
Ten.→Ins.	0.36	0.17	0.14	**0.57**	0.26	0.38
EnHi→FrJUDGE	0.47	-0.17	0.03	0.56	0.23	**0.59**
FrJUDGE→FrJUDGE	0.47	-0.17	0.03	**0.60**	0.22	0.48

Table V
E4: SANITY CHECKS (MEAN $\hat{y}$; SHARE WITHIN TOLERANCE)

System	Ident.	≥ 9	Unrel.	≤ 2
chrF	10.00	100%	2.89	4%
Embedding cosine	7.10	0%	4.36	20%
Token-align. F1	7.15	0%	5.20	0%
LegalDiff-GBR	9.99	100%	1.22	94%
ML-JudgeBERT (text)	9.41	80%	4.78	19%
ML-JudgeBERT+LegalDiff	9.98	100%	1.70	82%
ML-JudgeBERT+LegalDiff, no aug.	9.92	99%	1.88	74%

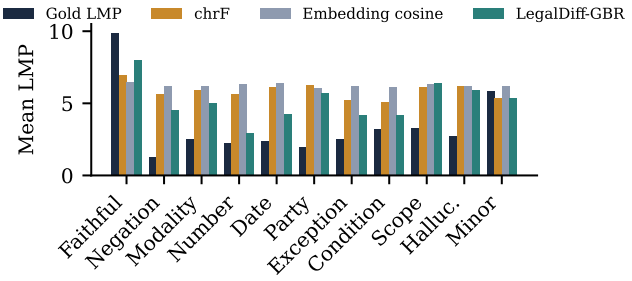


Figure 1. E5: mean LMP by gold error type (out-of-fold predictions, EN+HI).

every slice (e.g. English: 0.41 vs. 0.42 for token alignment; HI→HI: 0.34 vs. 0.37 for chrF). Generic similarity is especially unreliable for Hindi, where English-centric subword vocabularies degrade to characters.

Why does the encoder underperform? JUDGEBERT fine-tunes an encoder that already knows the language; our encoder had to learn English, Hindi and legal semantics from 256 pairs. Its weak standalone result is evidence for pretraining, not against cross-encoding: once given LegalDiff it becomes competitive and is the best system on zero-shot FrJUDGE.

IX. LIMITATIONS

(1) Gold scores are rubric-based silver labels written by the authors; inter-annotator agreement has not yet been measured, so absolute numbers may be optimistic, especially because LegalDiff's categories mirror the error taxonomy. The FrJUDGE results, based on independent human labels, partly address this. (2) Each corrupted pair contains one controlled error; naturally produced simplifications may combine several. (3) The encoder was trained from scratch for lack of pretrained checkpoints. (4) LegalDiff's lexicons are hand-built for English and Hindi, with a small French list (74 entries) added only for the FrJUDGE check. (5) Evaluation is sentence-level without document context.

Table VI
E5: ERROR-TYPE EXPLANATION QUALITY (LEGALDIFF-GBR SCORES)

Type	n	P	R	F1
NONE	160	0.82	0.69	0.75
SCOPE	22	0.00	0.00	0.00
DATE	20	0.61	0.55	0.58
OMIT_COND	20	0.32	0.30	0.31
ENTITY	16	0.37	0.81	0.51
HALLUC	16	0.18	0.12	0.15
MINOR	16	0.23	0.44	0.30
MODALITY	16	0.43	0.56	0.49
NEGATION	14	0.20	0.50	0.29
EXC_REMOVED	12	1.00	0.50	0.67
NUMBER	8	0.75	0.38	0.50

X. CONCLUSION AND FUTURE WORK

We extended JUDGEBERT's legal meaning preservation task to English and Hindi and to tenancy law, released a 320-pair parallel dataset with typed legal errors and sanity sets, and showed that generic similarity is not a legal-meaning metric ($r \leq 0.37$). Explicit, bilingual legal-difference modelling gives the strongest and most transferable signal ($r = 0.60$ in-domain; 0.56 zero-shot on human-labelled French data) and produces typed explanations (error-detection F1 0.79). Next steps are double human annotation with Krippendorff's α, fine-tuning XLM-R and MuRIL backbones on GPU, LLM-generated simplifications with multiple errors, and paragraph-level context.

ACKNOWLEDGMENT

We thank Dr. Surabhi Solanki for guidance, and the JUDGEBERT authors for releasing FrJUDGE and their code.

REFERENCES

[1] D. Beauchemin, M. Albert-Rochette, R. Khoury, and P.-L. Déziel, "JUDGEBERT: Assessing legal meaning preservation between sentences," in *Proc. EMNLP*, 2025, pp. 92–118.

Table VII
E5: QUALITATIVE ANALYSIS OF 18 LEGALLY DIFFICULT CASES (SCORES ON THE 1–10 LMP SCALE; $^{\times}$ = WRONG PREDICTED TYPE)

#	Lang	Gold type	Legal change (original → simplification)	Gold	chrF	Emb.	GBR	Hybrid	Predicted type
1	HI	MODALITY	"must not" use for hire unless covered → "may" use unless excluded	1	6.2	7.1	4.2	9.2	SCOPE$^{\times}$
2	HI	ENTITY	benefit goes to nominee on policyholder's death → roles swapped	1	7.6	7.0	5.2	6.2	ENTITY
3	HI	NEGATION	landlord shall not cut water/power → may cut if rent unpaid	1	7.2	7.0	4.5	9.3	NEGATION
4	HI	OMIT_COND	entry after 24-h notice, 8 a.m.–8 p.m. → any time, no notice	1	4.2	6.9	2.7	1.3	NEGATION$^{\times}$
5	HI	HALLUC	eviction only by legal procedure → after 7 days' notice	1	5.0	6.8	6.0	1.4	ENTITY$^{\times}$
6	EN	ENTITY	benefit goes to nominee on policyholder's death → roles swapped	1	8.3	6.5	7.9	9.9	NONE$^{\times}$
7	EN	NEGATION	"does not cover" wear and tear → "pays for" it	1	8.8	6.1	8.9	9.9	NONE$^{\times}$
8	HI	DATE	newborn covered from 91st day → from day of birth	2	5.6	7.1	5.1	8.1	DATE
9	HI	EXC_REMOVED	liable for damage, wear and tear excepted → including wear and tear	2	5.7	7.0	5.4	5.3	EXC_REMOVED
10	HI	SCOPE	insurer pays rateable share → pays entire loss	2	5.4	7.0	5.9	9.1	MINOR$^{\times}$
11	HI	NUMBER	deposit $\leq$ 2 months' rent → $\leq$ 6 months' rent	2	4.9	6.9	2.3	1.1	DATE$^{\times}$
12	EN	DATE	six-month lock-in → three-month lock-in	2	7.2	6.6	5.5	10.0	DATE
13	EN	NUMBER	deductible Rs 5,000 → Rs 50,000	2	8.4	6.6	5.3	9.4	NUMBER
14	EN	OMIT_COND	entry after 24-h notice, 8 a.m.–8 p.m. → any time, no notice	1	4.6	5.4	3.5	1.1	DATE$^{\times}$
15	EN	HALLUC	eviction only by legal procedure → after 7 days' notice	1	5.0	5.4	3.6	1.0	MODALITY$^{\times}$
16	EN	EXC_REMOVED	no hazardous goods except cooking gas → including cooking gas	2	7.7	6.1	7.6	9.8	NONE$^{\times}$
17	HI	NONE	faithful: rent +5% after every 11 months	10	2.0	6.2	1.8	5.4	HALLUC$^{\times}$
18	EN	NONE	faithful: rent +5% after every 11 months	10	3.2	5.2	9.0	7.1	NONE

[2] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. ACL*, 2002, pp. 311–318.

[3] M. Popović, "chrF: Character n-gram F-score for automatic MT evaluation," in *Proc. WMT*, 2015, pp. 392–395.

[4] W. Xu, C. Napoles, E. Pavlick, Q. Chen, and C. Callison-Burch, "Optimizing statistical machine translation for text simplification," *TACL*, vol. 4, pp. 401–415, 2016.

[5] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.

[6] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating text generation with BERT," in *Proc. ICLR*, 2020.

[7] D. Beauchemin, H. Saggion, and R. Khoury, "MeaningBERT: Assessing meaning preservation between sentences," *Frontiers in Artificial Intelligence*, vol. 6, 2023.

[8] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, "G-Eval: NLG evaluation using GPT-4 with better human alignment," in *Proc. EMNLP*, 2023, pp. 2511–2522.

[9] A. Conneau *et al.*, "Unsupervised cross-lingual representation learning at scale," in *Proc. ACL*, 2020, pp. 8440–8451.

[10] S. Khanuja *et al.*, "MuRIL: Multilingual representations for Indian languages," arXiv:2103.10730, 2021.

[11] K. Krippendorff, *Content Analysis: An Introduction to Its Methodology*, 4th ed. Thousand Oaks, CA: SAGE, 2018.